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Book Sales Performance & BI Reporting Design

Part 1: Analysis & Visualization

Excel Visualizations

See the full workbook here:

<https://docs.google.com/spreadsheets/d/135hCLc5bzTbqpyHmko29IUGWviMFhuOW/edit?usp=sharing>

for the four supporting charts: Monthly Revenue by Title, Title A Conversion Rate Trend, Total Revenue by Department, and Title D Monthly Units.

Findings & Interpretation

Looking at the data by book category, Adult titles bring in the most total revenue at \$8.5 million, which makes sense once you consider Adult books likely carry a higher price per copy. Kids titles come in second for revenue at \$4.1 million, but they actually outsell everything else in raw units, and Kids also has the best conversion rate of the three categories at 17.5%. The pattern is that Adult books earn more per sale while Kids books move higher volume at a lower price point.

At the individual title level, Title A sells the most units of any title, Title B brings in the most revenue, and Title A leads in conversion rate at 27.8% (averaging its monthly rates). Title A is the most interesting case in the whole dataset. Its unit sales tell a clear story over time: 274,030 units in 2023, dropping to 83,167 in 2024, and down to 47,326 in 2025. At the same time, its conversion rate has been climbing the entire time, from single digits back in 2023 up to nearly 50% by 2026. Put together, this looks like a title that had a huge launch spike in 2023 driven by heavy traffic, and has since settled into a smaller but more efficient audience.

Title D also shows negative units and revenue in June and July 2025 (-195 units / -\$10,018, and -1,471 units / -\$40,005). That means returns outpaced new sales those two months. It's worth confirming with Finance or Sales before assuming it's a trend. It could be a bulk return, a recall, or a data entry issue.

Part 2: Business Intelligence & Reporting

BI Structure: Dimensions, KPIs, and Filters

Dimensions: Title, Department, and Date (with Year, Quarter, and Month as a drill-down hierarchy so users can click from a yearly view down into individual months).

KPIs: Total Revenue, Total Units Sold, Average Conversion Rate, and Total Traffic as the headline numbers. Supporting metrics like best-performing month or best-performing year would live in a secondary section for anyone who wants to dig deeper. (Average Conversion Rate here uses the same method as Part 1 — averaging monthly rates, not total units divided by total traffic — since the two methods can produce different results.)

Filters: Year, Month, Department, and Title, so a user can narrow the whole report down to a specific slice, like just Kids titles in 2024, or just Title A across all time.

Layout: I'd design the main page to follow natural left-to-right, top-to-bottom eye movement (a Z-pattern), with the most important numbers in the top-left. The top row would hold the headline KPI cards: Total Revenue, Total Units Sold, Average Conversion Rate, and Total Traffic. Below that, I'd place Total Revenue by Department next to Monthly Revenue by Title, and under those, Conversion Rate Trend next to Monthly Units. Filters for Year, Month, Department, and Title would sit at the top of the page so they apply to every chart at once. I'd then build separate tabs for deeper dives into each metric: one for Monthly Revenue by Title, one for Conversion Rate Trend, one for Total Revenue by Department, and one for Monthly Units.

Tailoring the Experience: Finance vs. Marketing

I'd build tailored views for each audience rather than one dashboard for everyone, since Finance and Marketing care about different things even though they're looking at the same data. I'd do this as two separate report pages within the same Power BI file, both pulling from one shared data model. That way, each team gets a view built around what matters to them, but the numbers never conflict, since they're both coming from the same source underneath.

Finance: I'd lead with Total Revenue, broken down by Department and Title, along with trends by Month and Year. Since Finance cares most about accuracy, I'd also surface any anomalies automatically, like Title D's negative revenue in mid-2025, where returns outpaced sales. That's the kind of thing Finance would want flagged immediately, not buried in a chart they'd have to go looking for.

Marketing: I'd lead with Traffic, Conversion Rate, and POS Units, since those tie directly to campaign performance and audience behavior. I'd highlight the highest-traffic months so Marketing can connect spikes to specific launches or promotions, and also flag the lowest-performing months so they can investigate whether it's a seasonal dip or a real problem with a campaign. Title A is a good example of this. Its traffic dropped sharply after 2023 while conversion kept climbing, which tells a different story than raw traffic alone would.

DAX Measures

Total Revenue: = SUM(Table[Revenue])

YTD Revenue: = TOTALYTD(SUM(Table[Revenue]), Table[Date])

Sales Velocity: = DIVIDE(SUM(Table[POS Units]), DISTINCTCOUNT(Table[Month]))

This measures units sold per month, since the data was provided monthly. I'd confirm the intended time unit (week vs. month) with the business, since "velocity" can mean different things depending on the context.

Part 3: Problem Solving

Scenario: Power BI Numbers Don't Match Finance's Numbers

I would investigate by breaking the mismatch down into parts instead of trying to solve the whole gap at once, checking by title, by department, and by month, one at a time. I'd start by looking at the number of units sold and when they were logged, to check for issues with how or when the data was recorded. One likely cause is timing on returns and refunds: if Power BI counts a return in the month it happened but Finance counts it in the month it was processed, the two systems would show different numbers for that period. Other possible causes could include rounding differences, decimal errors, or currency mismatches. I would also look at Finance's total revenue numbers and work backward to trace which months and departments the difference is coming from.

While it's unresolved, I would tell the manager what I believe the problem is as I find it, give regular updates and let him know immediately if I find the cause, and prioritize any feedback he gives me along the way so there are no surprises.

Scenario: Diagnosing a Slow-Loading Report

I would first narrow down exactly where the slowness is happening, whether it's one specific page, a certain time of day, or the whole report. I'd also look at how the report is built, checking for things like how much data is being loaded at once and how many visuals are on a single page, and isolate the problem by testing one part at a time to see what's actually causing the slowdown.

Once I know the cause, likely fixes would be filtering out unnecessary data instead of loading everything since 2023, and splitting a busy page with too many visuals into two or three simpler pages.

As a quick fix while working on the real solution, I would set a default filter so the report opens already showing just the current year instead of all four years of data at once. This gets users a faster first impression right away, even before the deeper fixes are in place.

Scenario: Managing a Large Dashboard Request

I would start by asking the Director which metrics matter most to him, prioritizing sales-related KPIs and any marketing metrics that could tie back to sales. I'd also let him know that time is a real constraint, and recommend starting with a smaller set of 5-8 KPIs that matter most, since a dashboard with so many numbers may be hard for anyone to actually read at a glance, and I can always expand it later if he or anyone else still wants more. For slicers, I'd prioritize time-based ones like month and day, since those are most useful for tracking sales trends. For the daily email exports, Power BI has a built-in scheduled export feature, so I'd set that up to automatically send the key KPIs on a daily schedule rather than building something custom.

Full Excel workbook:

<https://docs.google.com/spreadsheets/d/135hCLc5bzTbqpyHmko29IUGWviMFhuOW/edit?usp=sharing>